

No. of Functional Features Included in the Solution

Date	30 September 2026
Team ID	SWTID-2026-3580
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Student Q&A;	Answers student questions using Gemini-generated responses.	qna.py / /qa	Done	0.75
2	Topic Explanation	Explains a topic in a simpler, student-friendly form.	explanation_module.py / /explain	Done	0.75
3	Quiz Generation	Generates multiple-choice quiz questions with options and answers.	quiz_module.py / /quiz	Done	0.75
4	Long-Passage Summarization	Summarizes lengthy study content into concise output.	summary_module.py / /summarize	Done	0.50
5	Learning Recommendations	Provides beginner, intermediate and advanced learning guidance.	learning_path.py / /learn/recommendations	Done	0.75
6	Web Learning Interface	Provides task selection, input forms, results, and quiz answer checking.	templates/index.html + static/style.css	Done	0.50
7	Gemini API Integration	Connects the backend to Google Gemini through a centralized client module.	gemini_client.py	Done	0.50
8	Local Model Support	Uses LaMini-Flan-T5-783M for local concept explanation.	explanation_module.py	Done	0.50

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5

Additional / Nice-to-Have Features	Responsive visual styling, animated UI effects, reduced-motion support, and backward-compatible API routes.
Features Tested & Verified	Health endpoint, validation, Q&A,, summary, quiz, and learning-recommendation flows are covered by the project test suite; frontend task flows were checked during functional testing.

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	1	Web learning interface, task selection, input forms, results and quiz answer checking.
2	Backend / Logic	5	Q&A,, explanation, quiz generation, summarization and learning recommendations.
3	Database / Storage	1	Application configuration and local model files; no persistent database in the current MVP.
4	API / Integration	1	Google Gemini API through gemini_client.py and FastAPI REST endpoints.
5	Security / Authentication	1	GEMINI_API_KEY is loaded from environment configuration; no user-login system in the current MVP.